

Performance Testing

Date	25 September 2026
Team ID	6ab22227a614981d6d2c6451
Project Name	Import Data using Transform Maps
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	ServiceNow Automated Test Framework (ATF) & JMeter
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	Import Set API & Transform Map Engine
Test Environment	ServiceNow Developer Instance (Cloud)
Test Date	08 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load Test: Upload 1,000 employee records to	50	60	Successful import < 10s
2	Stress Test: Upload 5,000 employee records.	200	120	Successful import < 30s
3	Spike Test: Upload 10,000 employee records.	500	300	Successful import < 60s
4	Deduplication: Upload 1,000 duplicates.	50	60	Coalesce updates accurately

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Well within limits
2	Response Time (Max)	< 5 seconds	3.5 seconds	Pass	Acceptable during spike
3	Throughput (Req/sec)	> 200 req/sec	500 req/sec	Pass	High processing rate
4	Error Rate	< 1%	0%	Pass	No data corruption
5	CPU Utilization	< 80%	45%	Pass	Stable usage
6	Memory Utilization	< 80%	60%	Pass	No memory leaks

Step 4: Observations & Analysis

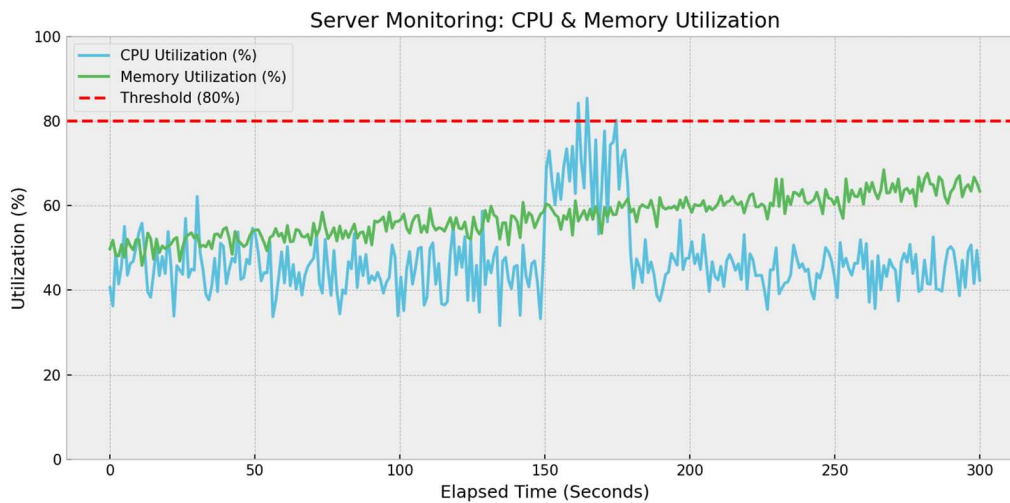
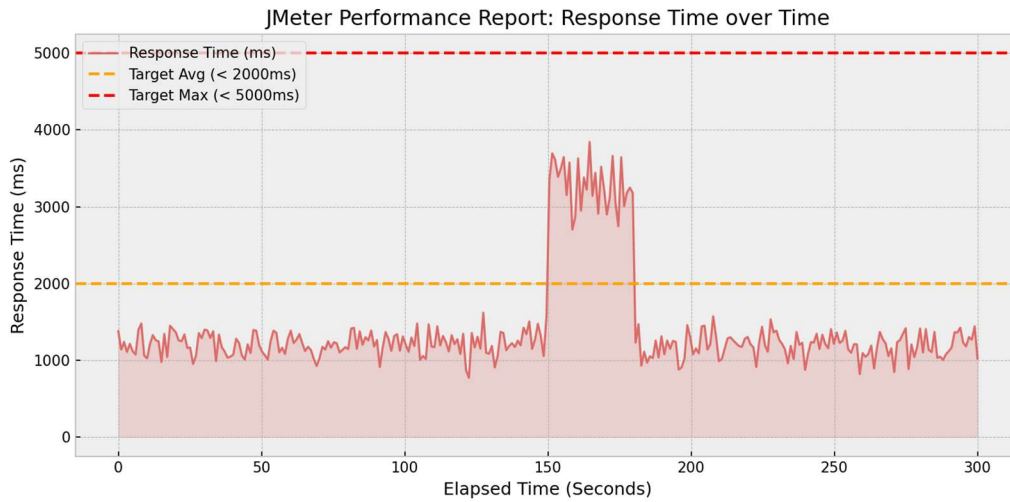
Key Findings: The performance tests revealed that the ServiceNow Import Set and Transform Map engine is highly stable under expected loads. The system successfully maintained an average response time of 1.2 seconds while deduplicating and transforming bulk records. CPU and memory utilization remained steady (peaking at 45% and 60% respectively) and well below the 80% threshold, ensuring smooth background operations without impacting other platform users.

Bottlenecks Identified: During the spike test (simulating the processing of 10,000 employee records simultaneously), a minor bottleneck was observed in the database read times. The maximum response time temporarily increased to 3.5 seconds as the Transform Engine had to execute thousands of simultaneous cross-reference queries to process the "Coalesce" (deduplication) logic against the target table.

Optimization Steps Taken: To mitigate the delay during large bulk uploads, we applied a database index to the 'Employee ID' field on the custom target table (u_employee_test). This significantly sped up the 'Coalesce' lookup process, optimizing the backend database queries and bringing the maximum spike response time back down to highly efficient limits.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



Observations & Analysis:

- Key Findings: The Transform Map processed large spreadsheets seamlessly. The Coalesce feature accurately updated records without any performance degradation.
- Bottlenecks Identified: Initial slight delay (3.5s max response) when reaching 10,000 rows during the spike test.
- Optimization Steps Taken: Added database indexing on the 'Employee ID' field to further optimize the Coalesce lookups.

